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Incumbents Adaptation to Competence-Destroying Change: Role of Prior Experience and Knowledge Sourcing

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For the past several decades, the topic of new technological developments has been of interest to strategy scholars (Ansoff and Stewart, 1967; Cooper and Schendel, 1976; Rosenbloom and Cusamano, 1987; Rothaermel, 2001a, 2001b; Tushman and Anderson, 1986). Scholars have repeatedly focused on this topic as innovation and new product development are critical for a firm's survival and growth (Penrose, 1995) and have stated that one of the core competencies needed to develop new technologies is the ability to assimilate and recombine knowledge in unique combinations (Rosenkopf and Nerkar, 2001). Simply put, successful innovation and new product development require the ability to create and use new knowledge to offer novel products or services to customers. Research indicates that firms who possess prior experience are in a better position to innovate (Carroll *et al.*, 1996; Henderson and Cockburn, 1994; Klepper and Simons 2000; Nerkar and Roberts, 2004). Prior studies have also highlighted the importance of knowledge and experience in enabling firms to successfully adapt to changes in technology (Bartel and Lichtenberg, 1987, 1991; Siegel, 1999; Siegel *et al.*, 1997). This stream of literature specifically states that a firm's technological and product-market experience enables it to "combine knowledge elements into valuable new products" (Nerkar and Roberts, 2004). Although this research stream makes valuable contributions to the innovation literature, it provides limited insights on the usefulness of prior

experience during technological change, especially those which are competence-destroying in nature. Tushman and Anderson (1986) stated that competence-destroying technological change creates a new class of products which "...require new skills, abilities, and knowledge in both the development and production of the product. The hallmark of competence-destroying discontinuities is that mastery of the new technology fundamentally alters the set of relevant competences within a product class" (Tushman and Anderson, 1986: 442). In the event of competence-destroying change, the existing products are replaced by new products based on the new technologies. For example, the photography industry experienced competence-destroying change when the technology changed from chemical-based to digital technologies giving rise to new products (Tripsas and Gavetti, 2000).

However, recent research finds that incumbent firms are not only capable of surviving competence-destroying changes but are also able to thrive under them (Tripsas, 1997; Rothaermel, 2001a, 2001b; Nicholls-Nixon and Woo, 2003). Traditionally, strategy scholars have traced the ability of firms to adapt and survive in changing environments to the capabilities and stocks of resources and knowledge endowed in the firms (Nicholls-Nixon and Woo, 2003; Tripsas, 1997) and their ability to collaborate with emerging technology firms (Rothaermel, 2001a, 2001b). For example, traditional pharmaceutical firms' distribution channels and manufacturing capabilities give them the ability to partner with biotechnology firms and learn the new technology. However, much of the literature on incumbent's adaptation to technological change primarily focuses on firm-level variables (e.g., alliances undertaken by a firm). While prior studies have explored relationships between prior competencies and research productivity (i.e., patents) (e.g., Henderson and Cockburn, 1994) and the influence of prior experience on product development (Marsh and Stock, 2006; Nerkar and Roberts, 2004), there is a dearth of studies investigating this relationship in the context of technological change (e.g., Sosa, 2009). This study seeks to fill the gap in the existing literature by examining pharmaceutical firms' adaptation to biotechnology at a firm-therapeutic level (e.g., Merck-Oncology).

The next section briefly reviews the organizational learning literature and comments on how organizational learning enables incumbent firms to develop new capabilities needed to adapt to technological change. Section 3 develops the hypotheses relating the incumbents' prior experience and knowledge sourcing to incumbents' adaptation success. Section 4 discusses the methodology and data, and subsequently the results. Empirical evidence from a sample of 1,932 alliances between incumbent pharmaceutical firms and new biotechnology firms reveals that incumbent firms that leverage their prior experience, develop absorptive capacity, and undertake external boundary spanning search are successful in adapting to the new technologies. The study concludes with a discussion of the findings, managerial implications, and future research opportunities.

THEORETICAL OVERVIEW

A firm must undertake learning during periods of technological change in order to develop new capabilities and adapt to the new technological paradigm. According to organizational learning theory, knowledge acquisition is the key aspect of organizational learning, including drawing on extant knowledge and enhancing the existing knowledge base within the firm through environmental scanning and organizational experience (Argote, 1999; Grant, 1996; Uhlenbruck *et al.*, 2003). Learning from internal and external sources has been viewed as one of the most prominent competencies for firms

competing in knowledge-intensive domains (Mitchell and Singh, 1996; Nicholls-Nixon and Woo, 2003; Rothaermel and Alexandre, 2008). The critical role played by internal and external learning during adaptation to technological change has been highlighted by prior literature, which is discussed in more detail below.

Internal Learning

Technological change affects all incumbents due to the high costs and uncertainty associated with technological discontinuities. However, despite these challenges, the process of internal learning is one of the most effective ways for a firm to upgrade and build competences (Cohen and Levinthal, 1990). One aspect of learning that is key to knowledge acquisition is that of absorptive capacity. Cohen and Levinthal (1989, 1990) define absorptive capacity as the ability to appraise, assimilate, and apply outside knowledge to commercialize new products. They contend that an essential condition for a firm's successful exploitation of external knowledge is development within the firm of the ability to absorb such capabilities. Similarly, Rosenberg asserts that internal R&D is needed because it takes "a substantial research capability to understand, interpret, and appraise knowledge that has been placed upon the shelf" (1990: 171). Furthermore, Gambardella (1992) stresses this point by observing that large pharmaceutical firms with better internal basic research programs are in a better position to exploit external knowledge. This suggests that failure to invest in learning can prevent a firm from keeping up with technological progress in an industry (Schilling, 1998).

The significance of internal development is also emphasized by Kogut and Zander's (1992) study in which the concept of combinative capabilities was introduced. They describe combinative capabilities as a function of a firm's extant knowledge and its internal and external learning experiences. They state that new learning such as innovations is a product of a firm's combinative capabilities to generate new applications from existing knowledge. Recent research also suggests that a firm's level of absorptive capacity is influenced by its combinative capabilities (Bosch *et al.*, 1999). For example, knowledge of the traditional drug development process may be inadequate for developing biotechnology-based drugs. Pharmaceutical firms have to undertake learning in order to develop the required absorptive capacity and capabilities needed to build biotechnology-based drugs, which is a highly complex and difficult task. However, not all firm-specific knowledge is destroyed during competence-destroying change. A recent study by Sosa (2009) indicates that application-specific knowledge could still be valuable in new technological domains. This application-specific knowledge along with knowledge of downstream activities could facilitate new knowledge development through internal and external learning.

External Sourcing

Complementarily, there also exists a substantial body of literature which suggests that, when the technological basis for competition in an industry is changing, it is essential for incumbent firms to establish strategic alliances in order to develop expertise in new and emerging technologies (Burgelman and Rosenbloom, 1989; Hoang and Rothaermel, 2005; Rothaermel, 2001a, 2001b). External relationships can also be used to anticipate and manage the adjustment to shifts in the prevailing technological paradigm. Researchers have argued that learning or acquisition of new capabilities from partner firms is one of the strongest motives for forming strategic alliances (Mody,

1993; Powell and Brantley, 1992). Alliances provide access to valuable information and knowledge to the collaborating firms (Arora and Gambardella, 1990; Powell *et al.*, 1996; Teece, 1992) as well as opportunities for firms to absorb know-how or skills and develop new capabilities. During periods of technological change, firms are driven to reconstruct their “knowledge network” - an organizational arrangement in which the firm collaborates with a group of external organizations to develop knowledge as the locus of innovation will be found in “networks of learning”, rather than in individual firms (Powell *et al.*, 1996).

This is the situation that confronted incumbent firms in the pharmaceutical industry with the emergence of biotechnology. In this industry, the established firms were forced to form relationships with new biotechnology firms in order to access the technologies needed to build their own capabilities in biotechnology (Rothaermel, 2001a, 2001b). Incumbent firms whose competitive positions are threatened by innovation use external relationships as a vehicle for adjusting their internal technological capabilities and accessing competencies and resources that they cannot produce internally in a timely fashion (Powell *et al.*, 1996; Powell, 1998). More recently, Rothaermel (2001a, 2001b) and his colleagues (Rothaermel and Deeds 2004; Rothaermel and Hill, 2005) investigate the influence external alliances have on incumbents’ ability to adapt to technological changes. In their studies, they find support for a positive relationship between alliance formation and new product development. They also find that extensive inter-firm cooperation between incumbents and new entrants during technological change leads to improvements in incumbent industry performance. Another important motivation behind incumbent firms’ decision to form alliances with emerging technology firms is to achieve an options approach to discovery (McGrath and MacMillan, 2000; McGrath and Nerkar, 2004). This allows incumbent firms to probe technologies being developed by multiple partners and avoid upfront commitment to a particular technology or partner. This is a viable strategy especially when the incumbent firms have control of complementary assets (e.g., drug approval process, marketing). Thus, strategic alliances help incumbent firms spread risk, share resources, increase speed of capability development, and minimize uncertainty by acquiring and exploiting knowledge developed by new technology firms (Hamel *et al.*, 1989; Folta, 1998).

In summary, previous studies provide insightful understanding of incumbents’ adaptation to technological changes. However, they predominantly provide a broad overview and do not look at the underlying intricacies behind the formation of each alliance. Looking at firm-therapeutic level reasoning for pursuing alliances could provide crucial information regarding incumbents’ adaptation to technological changes. Previous research has suggested that a firm’s competencies, its network of partners and capabilities of partnering firms has an impact on the degree of learning through alliances (e.g., Lane and Lubatkin, 1998; Mowery *et al.*, 1996; Powell *et al.*, 1996). This suggests that the benefits of an alliance do not accrue to a firm unless it has the capabilities and absorptive capacity to benefit from such an alliance.

HYPOTHESES

Prior Domain Experience

Prior experience not only enables firms to effectively deal with the challenges of the changing environment, but is also critical for opportunity recognition and innovation.

Numerous studies from the new product development literature highlight the role of prior experience in recognizing and developing opportunities (e.g., Carroll *et al.*, 1996; Holbrook *et al.*, 2000; Klepper, 2002; Leonard and Sensiper, 1998; O'Connor and Veryzer, 2001). In one of the earlier studies, Christensen and Peterson (1990) emphasize the importance of market and technology knowledge as prerequisites for recognizing opportunities. They view opportunity recognition as an innovation process and suggest that prior knowledge of market and technology enables entrepreneurs to identify potential opportunities. Similarly, Shane (2000) finds that prior experience can facilitate opportunity recognition. These include prior knowledge of ways to serve markets, prior knowledge of customer problems, and prior knowledge of markets. For example, Shane (2000) finds that an entrepreneur who has prior experience in the pharmaceutical industry knows how to serve a market that requires the product to be packaged in a way that would meet Food and Drug Administration (FDA) approval. Likewise, a study by Klepper and Simons (2000) shows that firms with experience in radio manufacturing are more likely to enter the TV industry and have better innovation performance and market share than companies that lack this experience. This suggests that prior knowledge in an area can enable one to identify a related entrepreneurial opportunity (Roberts, 1991). Researchers have also stated that repeated interaction with customers is beneficial for product innovation and have argued that prolonged exposure over time provides effective ideas for new product development (Leonard-Barton, 1992, 1995; Nambisan, 2002). By utilizing users' knowledge and integrating that knowledge back into product development, organizations can make more innovative products (Leonard-Barton, 1995; Von Hippel, 1988; Zirger and Maidique, 1990). In case of the pharmaceutical industry, the incumbent firms have knowledge to serve markets, solve customer problems, and manage regulatory issues. A firm's experience within product markets develops these complementary assets, which can be both technical, e.g., FDA approval process needed to commercialize drugs, knowledge about an application, etc. (Rothaermel, 2001a, 2001b; Sosa, 2009), and non-technical, e.g., marketing, sales and distribution assets needed to commercialize inventions (Abernathy and Clark, 1985). Researchers have further stated that complementary assets are a function of incumbents' prior experiences, are comprised of specific manufacturing capabilities, distributing channel experiences, sales, and service networks, and have a tacit component (Mitchell, 1989, 1991; Nerkar and Roberts, 2004; Teece, 1986; Tripsas, 1997). This makes the complementary assets highly unique and inimitable and this proves to be a source of valuable advantage to incumbents, as competitors cannot quickly amass such capabilities (complementary assets) due to time-compression diseconomies of scale (Dierickx and Cool, 1989). Furthermore, a recent study finds that application-specific R&D capabilities (i.e., knowledge about an application) can bestow incumbents with an advantage in adapting to competence-destroying change, as this specialized knowledge is accrued only to the incumbents (Sosa, 2009). Specific prior experience in a particular area gives the incumbent firm knowledge about applications, customers, distribution channels, and regulatory issues, all of which it can leverage to introduce products based on emerging technology. Thus, it is expected that specific prior experience (i.e., therapeutic area experience) will enhance success of adapting to the competence-destroying technology.

Hypothesis 1: The success of an incumbent in a therapeutic area is a positive function of its prior experience in that therapeutic area.

Boundary-spanning Search

According to the innovation studies, knowledge contributing to the development of innovations is regularly obtained from external sources (Gibbons and Johnston, 1974; Mansfield, 1995; Rothwell *et al.*, 1974; Von Hippel, 1988) as it is practically impossible for a firm to keep up with all the technological advances in its field exclusively through internal sourcing (Ettlie and Sethuraman, 2002; Hagedoorn, 1993; Rothaermel and Alexandre, 2008). Research also indicates that in high-technology industries, the locus of innovation is found within networks of learning rather than within a particular firm (Powell *et al.*, 1996). These networks can consist of diverse players such as competitors, universities, and government institutions that often collaborate and contribute to the technological advancement of the field.

As competence-destroying change typically involves knowledge that is novel to incumbent firms, the firms must undertake boundary-spanning search to access and learn this new knowledge (Nagarajan and Mitchell, 1998; Rosenkopf and Nerkar, 2001; Stuart and Podolny, 1996). For example, the emergence of biotechnology introduced a radical change in the drug discovery and development process, which prompted the incumbent pharmaceutical firms to learn this new technology by forming alliances with the biotechnology firms (Kenney, 1986; Rothaermel, 2001a, 2001b). Incumbent firms that undertake “external boundary-spanning” search (Rosenkopf and Nerkar, 2001) by forming alliances with other firms gain access to skills and knowledge. This leads to new opportunities and increases the incumbents’ chances for survival, growth, and innovation (Powell *et al.*, 1996; Silverman and Baum, 2002). A body of extant literature shows the positive effects of boundary-spanning search. For example, greater levels of R&D alliance participation increase patent output (Shan *et al.*, 1994) and result in more consequential innovations (Rosenkopf and Nerkar, 2001; Sorensen and Stuart, 2000). Similarly, innovation models (Nelson, 1982; Kortum, 1997) suggest that the extant knowledge base enables the search for new innovations by allowing firms to focus their search in the most fecund areas of opportunity (Fleming and Sorenson, 2004). This suggests that specific prior experience helps incumbent firms to identify potential opportunities and reposition themselves in the new technological domain.

Thus, this study expects that incumbent firms which form alliances (i.e., boundary-spanning search) in a particular therapeutic area with firms having the knowledge of the new technologies will enhance success of adapting to the new technology by developing new products based on the disruptive technology. Furthermore, it is expected that incumbent firms which conduct boundary-spanning search in familiar areas are more likely to attain success by developing new products based on the competence-destroying technology.

Hypothesis 2: The success of an incumbent in a therapeutic area is a positive function of its boundary-spanning search in a particular therapeutic area.

Hypothesis 3: The effect of boundary-spanning search on incumbent success is an increasing function of its prior experience.

Absorptive Capacity

Scholars have highlighted the importance of internal sourcing for creation of firm-specific knowledge that enables a firm to screen, evaluate, and take advantage of

external knowledge sources (Helfat, 1994; Nicholls-Nixon and Woo, 2003; Rothaermel and Alexandre, 2008). The importance of internal sourcing is highlighted by Cohen and Levinthal's (1989, 1990) seminal studies in which they suggest that absorptive capacity is a function of a firm's investment in internal R&D. This assertion is based on the argument that "while R&D obviously generates innovations, it also develops the firm's ability to identify, assimilate and exploit knowledge from the environment" (Cohen and Levinthal, 1989: 569). Moreover, in their reconceptualization of the absorptive capacity construct into potential and realized absorptive capacities, Zahra and George state that "potential capacity comprises knowledge acquisition and assimilation capabilities, and realized capacity centers on knowledge transformation and exploitation" (2002: 185). This suggests that growth of the organizational knowledge base depends on drawing connections between prior knowledge and new knowledge, such that absorptive capacity is needed by firms to assess, assimilate, and exploit this new internal and external knowledge (Cohen and Levinthal, 1990; Lane and Lubatkin, 1998).

In other words, a firm's absorptive capacity helps it link external and internal knowledge sources (Rothaermel and Alexandre, 2008), which allows the firm to make novel linkages among different types of knowledge (Rosenkopf and Nerkar, 2001). Furthermore, absorptive capacity enables knowledge recombination and integration activity, frequently leading to innovation (Kogut and Zander, 1992, 1996; Grant, 1996). When a firm possesses absorptive capacity, it not only becomes aware of the opportunities present in its environment, but also develops the capabilities to exploit these opportunities by combining internal and external sources of knowledge (Cohen and Levinthal, 1990). Prior research also indicates that firms with a higher level of absorptive capacity exhibit higher internal technological competence (Nicholls-Nixon, 1995; Rothaermel and Hill, 2005), which enables them to manage technological change effectively. Therefore, if incumbent firms want to increase their chances of survival in competence-destroying environment, they must develop absorptive capacity to position themselves to leverage their prior experience and also to take advantage of external knowledge sources. This means that those incumbent firms that take the initiative not only to learn the new technologies but also to develop the capabilities to further these new technologies will benefit the most from the technological discontinuity. Large absorptive capacity from inward investment in R&D can be expected to enable traditional pharmaceutical companies to better leverage their prior experience and also absorb more knowledge from alliances they form with biotechnology firms. Based on the above discussion, the following hypotheses are presented:

Hypothesis 4: The effect of prior experience on incumbent success is an increasing function of its absorptive capacity.

Hypothesis 5: The effect of boundary-spanning search on incumbent success is an increasing function of its absorptive capacity.

METHODS

Research Setting

This study assesses the ability of incumbent pharmaceutical firms to adapt to biotechnology by developing new drugs based on biotechnology science. The focal firms

are the incumbent pharmaceutical firms (e.g., Eli Lilly, Pfizer, etc.) as they attempt to adapt to the new drug development technology by forming collaborative partnerships (i.e., equity and non-equity alliances) with biotechnology firms (Amgen, Genentech, etc.). The unit of analysis is the firm-therapeutic level, i.e., the firm's alliance activities in a therapeutic area (for example, Merck's alliances in cardiovascular therapeutic area). The arguments are tested using panel data estimation techniques.

Sample and Data

The study identified 1932 alliances formed between incumbent pharmaceutical and new biotechnology firms in eighteen therapeutic areas by seventeen large pharmaceutical firms¹ during the time period from 1991 to 2004. The focal firms are traditional pharmaceutical firms listed under Standard Industrial Classification (SIC) codes 2834 and 5122, "Pharmaceutical Preparations" and "Wholesale-Drugs, Proprietaries & Druggists Sundries" respectively. In this industry, the incumbent pharmaceutical firms are primarily attempting to adapt to the emerging technologies by forming alliances (i.e., equity and non-equity partnerships) with new biotechnology firms (Rothaermel, 2000). This study utilized two primary sources: IMS Lifecycle database and Windhover RX Deals database, to gather information on alliances for both the pharmaceutical and the biotechnology partner over time. The main source for the new products information is IMS Lifecycle database and biotechnology industry publication (bio.org) and the primary source for alliance data is the Windhover RX Deals database. Since the late 1980s, the pharmaceutical industry has undergone reorganization due to mergers and acquisitions activity. This has a direct impact on the identities of the sample firms and subsequently on the data collection techniques used in the study. In this study, to control for mergers and acquisitions, the legally separate firms at the pre-merger stage were treated as a virtually combined firm. For instance, GlaxoSmithKline was formed in 2000. Pre-merger data on Glaxo Wellcome PLC and SmithKline Beecham were collected individually and then aggregated. This method makes the identities of sample firms consistent over the study period. Aggregation of firms to account for mergers and acquisitions should not be a severe problem and was previously used by other scholars (Rothaermel, 2001a). To create the alliance database, the study identified all pharmaceutical companies that were active as of 1991 by studying SIC listings and a variety of industry publication. Next, it constructed a database containing strategic alliances formed between incumbent pharmaceutical firms and new biotechnology firms from 1991 to 2004. The study accounted for acquisition or merger by creating a comprehensive "family tree" linking all companies in existence as of 2004 back to their various "ancestors" similar to Hoang and Rothaermel (2005). For example, two firms in the starting sample, Bristol-Myers Co. and Squibb, merged to form Bristol-Myers Squibb in 1989. In the analyses, the resulting organization was given the combined data of both companies, and the data were updated using the organization's new identity. An indicator variable was created to control for the effect of merger or acquisition on incumbent's performance.

¹The pharmaceutical industry is a concentrated industry consisting of a few large firms. Furthermore, since the mid-80s this industry has witnessed a series of acquisitions and mergers leading to further consolidations. Therefore, the sample of seventeen firms should cover the majority of the firms in this industry.

Variables and Measures

Dependent Variable. Adaptation to uncertain competence-destroying change was measured with a *count variable of biotechnology drugs* introduced by incumbent firms between 1991 and 2004 in a particular therapeutic area, i.e., this variable was measured at the firm-therapeutic level (e.g., Merck-Oncology). This measure is a good indicator of the incumbent firms' adaptation to the new technologies (Rothaermel, 2001a, 2001b). The biotechnology drugs information was obtained primarily from the biotechnology industry publication (bio.org). A pharmaceutical drug expert then classified each biotechnology drugs into the various therapeutic classes. The drug, Roferon-A, introduced by Hoffman-La Roche for chronic myelogenous leukemia, is an example of new biotechnology-based drug. The dependent variable was lagged by one-year to capture the delayed results of the incumbents' strategy.

Independent Variables. Researchers have stated that *prior experience* is beneficial for product innovation and have argued that prolonged exposure over time provides effective ideas for new product development (Leonard-Barton, 1995, Von Hippel, 1988). In the case of the pharmaceutical industry, the incumbent firms have prior experience and knowledge of ways to serve markets, customer problems, and regulatory issues. To measure this variable, the number of non-biotechnology drugs an incumbent has in a particular therapeutic domain each year was counted to capture the incumbent's experience in each specific area. It is argued here that an incumbent firm which has non-biotechnology drugs in a particular therapeutic domain will have the specific prior experiences needed to get approval (FDA) and market the drug in that specific product-market (i.e., therapeutic area). The primary source for this information was the IMS Lifecycle database.

Researchers have stated that incumbent pharmaceutical firms undertake alliances with biotechnology firms to gain access and learn new knowledge (Rothaermel, 2001a, 2001b). This *boundary-spanning search* activity enables the incumbent firm "to reposition itself technologically by moving away from local search" (Rosenkopf and Nerkar, 2001). The boundary-spanning search activity was measured by using a yearly count of alliances formed by the incumbent firms in each therapeutic area. This data was gathered from the Windhover database.

Following Cohen and Levinthal (1989, 1990), the incumbent's *absorptive capacity* was proxied by its R&D intensity, which is "Inflation Adjusted R&D" (in millions of dollars) by "Firm Size" (Number of Employees). The firm's R&D intensity was calculated by dividing the inflation-adjusted measure of research and development expenses by total number of employees for each year. The employee and financial data were gathered from annual reports of the firms and Compustat.

Control Variables. Based on prior studies, *firm size* is considered to approximate the financial resource position of pharmaceutical firms and may influence alliance activity (e.g., Burgers *et al.*, 1993; Gulati, 1995). The incumbent firm's size was controlled for by using a natural logarithm of the number of employees in each year. The data were gathered from annual reports and Compustat. Two different controls for acquisitions and mergers were used to control for spurious effects and confirm the robustness of the findings. A count of number of biotechnology acquisition (*Biotech Acquisitions*) an incumbent has made in a particular year was used to control for the effects of acquisitions of new biotechnology firms. A dummy variable was used to control for mergers and/or acquisition of traditional pharmaceutical companies (*Pharmaceutical M&A*). The

data were gathered from annual reports and the Windhover database. Similar to prior research, a count of number of *incumbent's patents* was used to control for the innovative capabilities of an incumbent. In the strategy literature, patents have widely been used as a measure of innovative capabilities of a firm (e.g., Rothaermel, 2001b). The patent data were coded according to the year of patent application, rather than the patent award date as there can be fairly long lags between patent applications and approvals (Ahuja, 2000a). By measuring application date, one is able to measure the innovation activity closer in time to when it was undertaken. The patent data were collected from the U.S. Patent and Trademark Office (CASSIS) database. Finally, dummy variables were included for each year to control for temporal effects.

DATA ANALYSIS AND RESULTS

The dependent variable of this study, technology adaptation, as measured by new biotechnology drugs, is a count variable. This variable is zero bounded and takes only positive integer values; therefore, a firm cannot have a negative count of new biotechnology drugs. Thus, the linear regression model's assumption of homoscedasticity, i.e., variance of the errors over the sample are similar, is violated. As the dependent variable is also a rare event, using ordinary least-squares regression on such data would provide unbiased estimation, but it will be less efficient than estimation approaches designed for limited dependent variables. A Poisson regression approach is designed specifically for limited dependent variables and provides more efficient estimation than ordinary least squares regression and therefore is more appropriate for such data (Hausman *et al.*, 1984; Henderson and Cockburn, 1996; Mezas, 2002). Consistent with prior innovation research, in this study, the Poisson regression model is utilized to test the hypotheses (e.g., Ahuja, 2000b; Ahuja and Katila, 2001; Stuart, 2000). Descriptive statistics and bivariate correlations are shown in Table 1 and the results of the analyses are depicted in Table 2.

Hypothesis 1 predicts a positive relationship between the success of an incumbent firm and its prior experience in a specific therapeutic area. According to Table 2, the coefficient of prior experience is positive and significant ($p < 0.01$), supporting Hypothesis 1. This suggests that incumbent firms' prior knowledge of the product market enables them to successfully adapt to the competence-destroying change. According to Hypothesis 2, there is a positive relationship between the success of an incumbent firm and the effect of boundary spanning search. As Model 2 in Table 2 shows, the coefficient of this variable is positive and significant ($p < 0.01$), providing support for this hypothesis. This indicates that incumbent firms who seek knowledge outside their organizational boundaries are successful in adapting to the competence-destroying change.

Hypotheses 3-5 explore the moderating effects of prior experience and absorptive capacity. Hypothesis 3 suggests that the effect of boundary-spanning search on incumbent success is an increasing function of its prior experience. According to Model 3 in Table 2, this hypothesis is supported ($p < 0.01$). This finding suggests that incumbents who seek novel knowledge from outside the firm in familiar product market areas (i.e., therapeutic areas) are successful in their adaptation efforts. Hypothesis 4 predicts a positive moderating role of absorptive capacity on the relationship between prior experience and incumbent success. According to Model 4 in Table 2, the coefficient of the interaction term is positive and significant ($p < 0.5$), providing support for Hypothesis 4. This finding indicates that higher levels of absorptive capacity positively

Table 1
Descriptive Statistics and Correlation Matrices

Variables	Mean	S.D.	1	2	3	4	5	6	7
Biotechnology Drugs	0.04	0.36							
Incumbent Patents	230.73	164.24	0.01						
Pharmaceutical M&A	0.11	0.31	-0.01	0.08**					
Biotech Acquisitions	1.22	1.52	0.03	0.32**	0.18**				
Firm Size	3.71	0.98	0.02	0.54**	0.14**	0.05*			
Absorptive Capacity	32.77	23.27	0	-0.13**	0.06**	-0.09**	-0.35*		
Prior Experience	0.28	0.68	0.11**	0.17**	0.01	0.10**	0.16	-0.04**	
Boundary-spanning search	0.84	1.22	0.06**	0.19**	-0.01	0.15**	0.08	-0.02	0.21

*p < 0.05; **p < 0.01

Table 2
Results of Fixed Effects Poisson Regression Analysis^a

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Control Variables</i>						
Incumbent Patents	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
Pharmaceutical M&A	-0.822+ (0.429)	-0.607 (0.433)	-0.638 (0.437)	-0.657 (0.430)	-0.582 (0.433)	-0.689 (0.435)
Biotech Acquisitions	0.248** (0.080)	0.207* (0.083)	0.225** (0.085)	0.186* (0.083)	0.217** (0.082)	0.204* (0.085)
Firm Size	0.811 (0.554)	0.975 (0.612)	1.138+ (0.622)	1.008+ (0.609)	0.760 (0.627)	0.945 (0.622)
<i>Independent Variables</i>						
Absorptive Capacity		-0.002 (0.011)	-0.001 (0.011)	-0.011 (0.012)	-0.018 (0.013)	-0.026+ (0.014)
Prior Experience		0.430** (0.090)	-0.050 (0.173)	-0.174 (0.283)	0.446** (0.092)	-0.524 (0.350)
Boundary-spanning search		0.185** (0.067)	0.003 (0.090)	0.174** (0.066)	-0.211 (0.166)	-0.498* (0.211)
<i>Interaction Effects</i>						
PriorExp x BoundarySearch			0.197** (0.053)			0.226** (0.054)
PriorExp x AbsCapacity				0.019* (0.008)		0.015 (0.010)
BoundarySearch x AbsCapacity					0.010** (0.004)	0.011** (0.004)
Log-likelihood	-344.085	-327.195	-320.072	-324.386	-323.394	-312.598
Wald χ^2	49.61	96.30	132.33	104.63	103.08	150.95

N = 1932. ^aStandard errors in parentheses. Year dummies are included but not reported for parsimony. +p < 0.10; *p < 0.05; **p < 0.01.

influence the relationship between prior experience and incumbent success. Hypothesis 5 posits that absorptive capacity positively moderates the relationship between boundary-spanning search and an incumbent firm's success. Model 5 in Table 2 shows that the coefficient of this variable is positive and significant ($p < 0.01$), supporting Hypothesis 5. This suggests that higher levels of absorptive capacity have a positive effect on the relationship between boundary-spanning search and an incumbent firm's success. Model 6 adds all the variables including controls, independent variables, and the interaction terms. Although the level of significance of some variables changed, the results shown in this model are consistent with the other models.

DISCUSSION AND CONCLUSION

The objective of this paper is to better understand incumbents' adaptation to competence-destroying technological change. Specifically, this paper investigates the relationship between incumbents' prior experience, boundary-spanning search, and absorptive capacity on their ability to adapt to the competence-destroying technological change. The support for the first hypothesis indicates the importance of leveraging prior experience in a particular therapeutic area during adaptation to technological change. Although previous research has generally highlighted the importance of prior experience in new product development (Marsh and Stock, 2006; Nerkar and Roberts, 2004), this relationship has relatively remained unexplored in the context of technological change. A noteworthy exception is a recent study by Sosa (2009), which explored the importance of prior application-specific knowledge during technological change. Specifically, Sosa (2009) finds that incumbent pharmaceutical firms that have prior experience in the oncology therapeutic area are at an advantage in the emerging biotechnology-based anticancer market. This suggests that prior oncology research knowledge is valuable and difficult for new entrants to develop, and incumbent firm's can leverage their prior knowledge of oncology to adapt to biotechnology. Although Sosa (2009) highlights the advantage of prior application-specific experience during technological change, their study focuses on only the oncology therapeutic area. In this study, a wider range of therapeutic areas are included, which not only validates the findings of Sosa (2009) but also leads to a more generalizable conclusion. In general, the results suggest that knowledge of the therapeutic area and the associated complementary assets (including knowledge about clinical and phase tests required for FDA approval in particular therapeutic area) enable the incumbent to adapt to the competence-destroying technological change.

The next hypothesis investigates the relationship between boundary-spanning search and incumbents' adaptation success in a particular therapeutic area. The support for the second hypothesis highlights the value of external boundary spanning search during competence-destroying change. This study finds that incumbents who explore knowledge by forming alliances with biotechnology firms are successful in adapting to competence-destroying technological change. Although the results of previous studies indicate the importance of external knowledge sourcing during competence-destroying change (Rothaermel, 2001a, 2001b), the level of analysis adopted in this study (i.e., firm-therapeutic level) allows it to extend the prior findings. The results suggest that developing collaborative partnerships in a particular therapeutic area with emerging technology firms (i.e., new network of partners) enables incumbent firms to adapt to the technological change. Specifically, the finding in this study is related to

the search for knowledge across firm boundaries, rather than search for knowledge across technological domains. The results also demonstrate that absorptive capacity is important for adaptation and has both direct and indirect effects on it. The interaction effects results indicate that absorptive capacity has a positive effect on the relationship between prior experience and adaptation success and also on the relationship between external boundary spanning search and adaptation success. Not only do the results underscore the positive moderating effects of absorptive capacity, they also illustrate the importance of internal generation of capabilities through investments in R&D.

Another contribution of this paper to the literature of innovation and technology adaptation is its research design and panel estimation techniques. Researchers called for more longitudinal studies to investigate and corroborate the causal claims made by prior studies (e.g., Drazin and Schoonhoven, 1996; Rothaermel 2001a, 2001b). This study takes a step in that direction and seeks to explain the complexities faced by incumbents attempting to adapt to competence-destroying changes. It is undertaken with the effort to present a clearer picture regarding the activities of the incumbents and the effect of those activities on adaptation. Moreover, this study provides key insights to managers regarding the importance of internal and external learning. The results indicate that firm managers should seek to develop firm's absorptive capacity as well as pursue external knowledge through boundary-spanning search during periods of competence-destroying change. More importantly, the results suggest that managers should seek to leverage the knowledge that is preserved in the organization during competence-destroying change (i.e., prior experience) as it has a positive impact on the incumbent's ability to adapt to the competence-destroying change. Although the findings of this paper allude to the advantages of prior experience, it is important that managers are cognizant about the downsides of over dependence on existing routines and capabilities. Studies have indicated that during periods of technological change the existing routines and capabilities can cause inertial tendencies, which hinder incumbents' response to technological change (Benner, 2009; Tripsas and Gavetti, 2000). However, managers that pay attention to emerging technologies in their industry can overcome this inertia and enter the emerging technological domains faster (Eggers and Kaplan, 2009).

The study presented here, like any other study, has its limitations and future research opportunities existing as a result of them. First, the findings of the study are limited to one industry, and hence may not be fully generalizable. Moreover, this study only investigates those alliances that occurred between 1991 and 2004. In addition, the study sample is restricted to only those alliances that appear in the Windhover database, possibly affecting the results of the study. In order to build on the findings of this study, future research should try to collect data from multiple different sources and also include other industries that experienced competence-destroying changes in order to improve the generalizability of the findings.

The single dependent variable measure (i.e., new biotechnology drugs introduced by incumbent firms) used in this study to capture incumbent firms' adaptation to competence-destroying change faces some issues. The drug approval process is a rare event and might not capture the full extent of learning and adaptation. Future researchers should use other measures such as R&D projects or drug trials in various phases to complement biotechnology drugs as a measure of adaptation success. Using multiple measures of the dependent variable will shed more light on the extent of the incumbents' adaptation to competence-destroying changes. Likewise, the measure for prior experience in a therapeutic area could be improved by using multiple measures for the prior experience variable such as research publications, phase trials, and patent

in each therapeutic area. Data availability limitations (i.e., use of secondary data) and data identification issues (e.g., difficulty in identifying biotechnology patents) prevent exploring the influence of prior experience on incumbents' adaptation in greater detail. Future research could enhance the findings of this study by utilizing both primary and secondary data sources to investigate the influence of prior experiences and other variables on incumbents' adaptation.

Another area that future research could explore is in understanding the effect of rigidities and cognitive inertia on incumbent adaptation at the firm-therapeutic level. A recent study by Eggers and Kaplan (2009) finds that managerial attention towards emerging technologies is associated with faster entry, and vice versa managerial attention towards existing technologies leads to slower response. Future research could investigate how various firm-therapeutic level factors (e.g., prior experience, core therapeutic area vs. non-core therapeutic area) influence managerial attention. A better understanding of the relationship between managerial attention and prior experience could assist managers in overcoming the dangers of myopia. Similarly, future research could explore how competitive-dynamics at the firm-therapeutic level affect incumbents response to competence-destroying change. Pursuing such questions will lead to a better understanding of factors that motivate or hinder incumbents to respond to technological change.

In conclusion, this study investigates incumbents' adaptation to competence-destroying change at the firm-therapeutic level and makes an important contribution to the technology and innovation literature. The results imply that incumbent firms should pursue knowledge development through both internal and external mechanisms and leverage prior experience in a particular therapeutic area to successfully adapt to competence-destroying change. However, much work remains to be done in this area. It is hoped that this study informs and motivates further work on this critical topic.

References

- Abernathy, W. J. and K. B. Clark. 1985. "Innovation: Mapping the Winds of Creative Destruction." *Research Policy* 14: 3-22.
- Ahuja, G. 2000a. "Collaboration Networks, Structural Holes, and Innovation: A Longitudinal Study." *Administrative Science Quarterly* 45: 425-455.
- _____. 2000b. "The Duality of Collaboration: Inducements and Opportunities in the Formation of Inter-Firm Linkages." *Strategic Management Journal* 21: 317-343.
- _____. and R. Katila. 2001. "Technological Acquisitions and the Innovation Performance of Acquiring Firms: A Longitudinal Study." *Strategic Management Journal* 22: 197-220.
- Ansoff, H. I. and J. M. Stewart. 1967. "Strategies for a Technology-based Business." *Harvard Business Review* (Nov-Dec): 71-83.
- Argote, L. 1999. *Organizational Learning: Creating, Retaining and Transferring Knowledge*. Boston, MA: Kluwer.
- Arora, A. and A. Gambardella. 1990. "Complementarity and External Linkages: The Strategies of the Large Firms in Biotechnology." *Journal of Industrial Economics* 38: 361-379.
- Bartel, A. P. and F. R. Lichtenberg. 1991. "The Age of Technology and Its Impact on

- Employee Wages." *Economics of Innovation and New Technology* 1: 215-31.
- _____ and _____. 1987. "The Comparative Advantage of Educated Workers in Implementing New Technology." *Review of Economics and Statistics* 69: 1-11.
- Benner, M. J. 2009. "Dynamic or Static Capabilities? Process Management Practices and Response to Technological Change." *The Journal of Product Innovation Management*. 26: 473-486.
- Bosch, F., H. Volberda, and M. Boer. 1999. "Coevolution of Firm Absorptive Capacity and Knowledge Environment: Organizational Forms and Combinative Capabilities." *Organization Science* 10: 551-568.
- Burgelman, R. and R. Rosenbloom. 1989. "Technology Strategy: An Evolutionary Process Perspective." *Research on Technological Innovation, Management and Policy* 4: 1-23.
- Burgers, W. P., C. W. L. Hill, and W. C. Kim. 1993. "A Theory of Global Strategic Alliances: The Case of the Global Auto Industry." *Strategic Management Journal* 14: 419-432.
- Carroll, G. R., L. S. Bigelow, M. D. Seidel, and L. B. Tsai. 1996. "The Fates of De Novo and De Alio Producers in the American Automobile Industry, 1885-1982." *Strategic Management Journal* 17: 117-38.
- Christensen, P. S. and R. Peterson. 1990. *Opportunity Identification: Mapping the Sources of New Venture Ideas*. (Paper presented at the 10th annual Babson Entrepreneurship Research Conference). Aarhus University Institute of Management: Denmark.
- Cohen, W. M. and D. A. Levinthal. 1990. "Absorptive Capacity: A New Perspective on Learning and Innovation." *Administration Science Quarterly* 35: 128-152.
- _____ and _____. 1989. "Innovation and Learning: Two Faces of R&D." *Economic Journal* 99: 569-596.
- Cooper, A. and D. Schendel. 1976. "Strategic Responses to Technological Threats." *Business Horizons* 19: 61-69.
- Dierickx, I. and K. Cool. 1989. "Asset Stock Accumulation and Sustainability of Competitive Advantage." *Management Science* 35: 1504-1514.
- Drazin, R. and C. B. Schoonhoven. 1996. "Community, Population, and Organization Effects on Innovation: A Multilevel Perspective." *Academy of Management Journal* 39: 1065-1083.
- Eggers, J. P. and S. Kaplan. 2009. "Cognition and Renewal: Comparing CEO and Organizational Effects on Incumbent Adaptation to Technical Change." *Organization Science* 20: 461-477.
- Ettlie, J. E. and K. Sethuraman. 2002. "Locus of Supply and Global Manufacturing." *International Journal of Operations & Production Management* 22: 349-370.
- Fleming, L. and O. Sorenson. 2004. "Science as a Map in Technological Search." *Strategic Management Journal* 25: 909-928.
- Folta, T. B. 1998. "Governance and Uncertainty: The Tradeoff between Administrative Control and Commitment." *Strategic Management Journal* 19: 1007-1028.
- Gambardella, A. 1992. "Competitive Advantages from In-House Basic Research: The U.S. Pharmaceutical Industry in the 1980s." *Research Policy* 21: 391-407.
- Gibbons, M. and R. Johnston. 1974. "The Role of Science in Technological Innovation." *Research Policy* 3: 220-242.
- Grant, R. M. 1996. "Toward a Knowledge-based Theory of the Firm." *Strategic Management Journal* 17: 109-122.
- Gulati, R. 1995. "Social Structure and Alliance Formation Patterns: A Longitudinal

- Analysis." *Administrative Science Quarterly* 40: 619-652.
- Hagedoorn, J. 1993. "Understanding the Rationale of Strategic Technology Partnering: Interorganizational Modes of Cooperation and Sectoral Differences." *Strategic Management Journal* 14: 371-385.
- Hamel, G., Y. L. Doz, and C. K. Prahalad. 1989. "Collaborate with Your Competitors - And Win." *Harvard Business Review* 67: 133-139.
- Hausman, J., B. Hall, and Z. Griliches. 1984. "Econometric Models of Count Data with an Application to the Patents-R&D Relationship." *Econometrica* 52: 909-938.
- Helfat, C. E. 1994. "Firm-Specificity in Corporate Applied R&D." *Organization Science* 5: 173-184.
- Henderson, R. and I. Cockburn. 1996. "Scale, Scope, and Spillovers: The Determinants of Research Productivity in Drug Discovery." *RAND Journal of Economics* 27: 32-59.
- _____ and _____. 1994. "Measuring Competence: Exploring Firm Effects in Pharmaceutical Research." *Strategic Management Journal* 15: 63-84.
- Hoang, H. and F. T. Rothaermel. 2005. "The Effect of General and Partner-specific Alliance Experience on Joint R&D Project Performance." *Academy of Management Journal* 48 (2): 332-345.
- Holbrook, D., W. M. Cohen, D. A. Hounshell, and S. Klepper. 2000. "The Nature, Sources, and Consequences of Firm Differences in the Early History of the Semiconductor Industry." *Strategic Management Journal* 21: 1017-1041.
- Kenney, M. 1986. *Biotechnology: The University-Industrial Complex*. New Haven, CT: Yale University Press.
- Klepper, S. 2002. "The Capabilities of New Firms and the Evolution of the US Automobile Industry." *Industrial and Corporate Change* 11 (4): 645-666.
- _____ and K. Simons. 2000. "Dominance by Birthright: Entry of Prior Radio Producers and Competitive Ramifications in the U.S. Television Receiver Industry." *Strategic Management Journal* 21: 997-1016.
- Kogut, B. and U. Zander. 1996. "What Firms Do? Coordination, Identity and Learning." *Organization Science* 7: 502-518.
- _____ and _____. 1992. "Knowledge of the Firm, Combinative Capabilities, and the Replication of Technology." *Organization Science* 3: 383-397.
- Kortum, S. S. 1997. "Research, Parenting and Technological Change." *Econometrica* 65: 1389-1420.
- Lane, P. J. and M. Lubatkin. 1998. "Relative Absorptive Capacity and Interorganizational Learning." *Strategic Management Journal* 19: 461-477.
- Leonard-Barton, D. 1995. *Wellsprings of Knowledge*. Boston, MA: Harvard Business School Press.
- _____. 1992. "Core Capabilities and Core Rigidities: A Paradox in Managing New Product Development." *Strategic Management Journal* 13: 111-125.
- Leonard, D. and S. Sensiper. 1998. "The Role of Tacit Knowledge in Group Innovation." *California Management Review* 40: 112-132.
- Mansfield, E. 1995. "Academic Research Underlying Industrial Innovations: Sources, Characteristics, and Financing." *The Review of Economics and Statistics* 77: 55-65.
- Marsh, S. J. and G. N. Stock. 2006. "Creating Dynamic Capability: The Role of Intertemporal Integration, Knowledge Retention, and Interpretation." *The Journal of Product Innovation Management* 23: 422-436.
- McGrath, R. G. and A. Nerkar. 2004. "Real Options Reasoning and a New Look at the R&D Investment Strategies of Pharmaceutical Firms." *Strategic Management Journal*

- 25: 1-21.
- _____ and I. C. MacMillan. 2000. "Assessing Technology Projects using Real Options Reasoning." *Research and Technology Management* 43: 35-49.
- Mezias, J. M. 2002. "Identifying Liabilities of Foreignness and Strategies to Minimize Their Effects: The Case of Labor Lawsuit Judgments in the United States." *Strategic Management Journal* 23: 229-244.
- Mitchell, W. 1991. "Dual Clock: Entry Order Influences on Incumbent and Newcomer Market Share and Survival When Specialized Assets Retain Their Value." *Strategic Management Journal* 12: 85-100.
- _____. 1989. "Whether and When: Probability and Timing of Incumbents' Entry into Emerging Industrial Subfields." *Administrative Science Quarterly* 34: 208-230.
- _____ and K. Singh. 1996. "Business Survival of Firms Using Hybrid Relationships in the American Hospital Software Systems Industry, 1961-1991." *Strategic Management Journal* 17: 169-195.
- Mody, A. 1993. "Learning Through Alliances." *Journal of Economic Behavior and Organization* 20: 151-170.
- Mowery, D., J. Oxley, and B. Silverman. 1996. "Strategic Alliances and Inter-firm Knowledge Transfer." *Strategic Management Journal* 17: 77-91.
- Nagarajan, A. and W. Mitchell. 1998. "Evolutionary Diffusion: Internal and External Methods Used to Acquire Encompassing, Complementary, and Incremental Technological Changes in the Lithotripsy Industry." *Strategic Management Journal* 19: 1063-1077.
- Nambisan, S. 2002. "Designing Virtual Customer Environments for New Product Development: Toward a Theory." *Academy of Management Review* 27: 392-413.
- Nelson, R. R. 1982. "The Role of Knowledge in R&D Efficiency." *The Quarterly Journal of Economics* 97: 453-70.
- Nerkar, A. and P. W. Roberts. 2004. "Technological and Product-Market Experience and the Success of New Product Introductions in the Pharmaceutical Industry." *Strategic Management Journal* 25: 779-799.
- Nicholls-Nixon, C. 1995. "Responding to Technological Change: Why Some Firms Do and Others Die." *Journal of High Technology Management Research* 6: 103-109.
- _____ and C. Woo. 2003. "Technology Sourcing and Output of Established Firms in a Regime of Encompassing Technological Change." *Strategic Management Journal* 24: 651-666.
- O'Connor, G. C. and R. W. Veryzer. 2001. "The Nature of Market Visioning for Technology-based Radical Innovation." *The Journal of Product Innovation Management* 18: 231-246.
- Penrose, E. T. 1995. *The Theory of the Growth of the Firm*. (3rd Ed). Oxford: Oxford University Press.
- Powell, W. 1998. "Learning from Collaboration: Knowledge and Networks in the Biotechnology and Pharmaceutical Industries." *California Management Review* 40: 228-240.
- _____ and P. Brantley. 1992. "Competitive Cooperation in Biotechnology: Learning Through Networks." In *Networks and Organizations: Structure, Form and Action*. Ed. N. Nohria and R. G. Eccles. Boston, MA: Harvard Business School Press.
- Powell, W., K. W. Koput, and L. Smith-Doerr. 1996. "Inter-Organizational Collaboration and the Locus of Innovation: Networks of Learning in Biotechnology." *Administrative Science Quarterly* 41: 116-145.

- Roberts, E. 1991. *Entrepreneurs in High Technology: Lessons from MIT and Beyond*. Oxford: Oxford University Press.
- Rosenberg, N. 1990. "Why Do Firms Do Basic Research (with Their Own Money)?" *Research Policy* 19: 165-174.
- Rosenbloom, R. S. and M. A. Cusamano. 1987. "Technological Pioneering and Competitive Advantage: The Birth of the VCR Industry." *California Management Review* 29: 51-76.
- Rosenkopf, L. and A. Nerkar. 2001. "Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disc Industry." *Strategic Management Journal* 22: 287-306.
- Rothaermel, F. T. 2001a. "Incumbent's Advantage through Exploiting Complementary Assets via Inter-Firm Cooperation." *Strategic Management Journal* 22: 687-699.
- _____. 2001b. "Complementary Assets, Strategic Alliances, and the Incumbent's Advantage: An Empirical Study of Industry and Firm Effects in the Biopharmaceutical Industry." *Research Policy* 30: 1235-1251.
- _____. 2000. "Technological Discontinuities and the Nature of Competition." *Technology Analysis and Strategic Management* 12: 149-160.
- _____ and M. Alexandre. 2008. "Ambidexterity in Technology Sourcing: The Moderating Role of Absorptive Capacity." *Organization Science* 20: 1-22.
- _____ and D. L. Deeds. 2004. "Exploration and Exploitation Alliances in Biotechnology: A System of New Product Development." *Strategic Management Journal* 25: 201-221.
- _____ and C. W. L. Hill. 2005. "Technological Discontinuities and Complementary Assets: A Longitudinal Study of Industry and Firm Performance." *Organization Science* 16: 52-70.
- Rothwell, R., C. Freeman, A. Horsley, V. T. P. Jarvis, A. B. Robertson, and J. Townshend. 1974. "SAPPHO Updated - Project SAPPHO Phase II." *Research Policy* 3: 258-291.
- Schilling, M. A. 1998. "Technological Lockout: An Integrative Model of the Economic and Strategic Factors Driving Technology Success and Failure." *Academy of Management Review* 23: 267-284.
- Shan, W., G. Walker, and B. Kogut. 1994. "Inter-Firm Cooperation and Startup Innovation in the Biotechnology Industry." *Strategic Management Journal* 15: 387-394.
- Shane, S. 2000. "Prior Knowledge and the Discovery of Entrepreneurial Opportunities." *Organization Science* 11: 448-469.
- Siegel, D. S. 1999. *Skill-Biased Technological Change: Evidence from a Firm-Level Survey*. Kalamazoo, MI: Upjohn Institute for Employment Research.
- _____, D. A. Waldman, and W. E. Youngdahl. 1997. "The Adoption of Advanced Manufacturing Technologies: Human Resource Management Implications." *Engineering Management* 44: 288-298.
- Silverman, B. S. and J. A. C. Baum. 2002. "Alliance-based Competitive Dynamics." *Academy of Management Journal* 45: 791-806.
- Sorenson, J. B. and T. E. Stuart. 2000. "Aging, Obsolescence and Organizational Innovation." *Administrative Science Quarterly* 45: 81-112.
- Sosa, M. L. 2009. "Application-Specific R&D Capabilities and the Advantage of Incumbents: Evidence from the Anticancer Drug Market." *Management Science* 55: 1409-1422.
- Stuart, T. E. 2000. "Interorganizational Alliances and the Performance of Firms: A

- Study of Growth and Innovation Rates in a High-Technology Industry." *Strategic Management Journal* 21: 791-811.
- _____ and J. M. Podolny. 1996. "Local Search and the Evolution of Technological Capabilities." *Strategic Management Journal* 17: 21-38.
- Teece, D. 1992. "Competition, Cooperation and Innovation." *Journal of Economic Behavior and Organization* 18: 1-25.
- _____. 1986. "Profiting from Technological Innovation: Implication for Integration, Collaboration, Licensing and Public Policy." *Research Policy* 15: 285-305. 1992.
- Tripsas, M. 1997. "Unraveling the Process of Creative Destruction: Complementary Assets and Incumbent Survival in the Typesetter Industry." *Strategic Management Journal* 18: 119-142.
- Tripsas, M. and G. Gavetti. 2000. "Capabilities, Cognition and Inertia: Evidence from Digital Imaging." *Strategic Management Journal* 21: 1147-1161.
- Tushman, M. and P. Anderson. 1986. "Technological Discontinuities and Organizational Environment." *Administrative Science Quarterly* 31: 439-465.
- Uhlenbruck, K., K. E. Meyer, and M. Hitt. 2003. "Organizational Transformation in Transition Economies: Resource-based and Organizational Learning Perspectives." *Journal of Management Studies* 40: 257-282.
- Von Hippel, E. 1988. *The Sources of Innovation*. New York, NY: Oxford University.
- Zahra, A. S. and G. George. 2002. "Absorptive Capacity: A Review, Reconceptualization and Extension." *Academy of Management Review* 27: 185-203.
- Zirger, B. and M. Maidique. 1990. "A Model of New Product Development: An Empirical Test." *Management Science* 36: 867-883.